Georgios Peikos

Applied AI Researcher & Engineer | Language Models | Information Retrieval

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Computer Science PhD and Assistant Professor at the **University of Milano-Bicocca**, specializing in **Natural Language Processing** and **Information Retrieval**. I design, build and evaluate language models and retrieval systems for domain-specific problems, from model training on HPC infrastructure to RAG systems and evaluation tools. My work combines hands-on research with technical team leadership and collaboration across academic and industry projects.

Selected Skills

Programming & Frameworks	Python | SQL | PyTorch | Hugging Face Transformers | scikit-learn
Language-Model Development	Training from scratch | Tokenizer training | Training data preparation | LoRA/PEFT | Full fine-tuning
Research & Evaluation	Benchmark development | Experimental evaluation | Error analysis | Statistical significance testing
Distributed Training	Megatron-LM | PyTorch DDP | Multi-GPU training | Slurm/HPC | Experiment tracking
Information Retrieval	BM25 | Text embeddings | Dense retrieval | Semantic search | Ranking | RAG
Engineering & Systems	AWS EC2/S3 | Docker | CI/CD | Git | UNIX/Linux
Leadership & Collaboration	Research team leadership | International collaboration | Mentoring
Languages	English (C2) | Italian (B1) | Greek (native)

Professional Experience

Assistant Professor (RTDA) — Information Retrieval and NLP 2024 – Present
University of Milano-Bicocca, Milano, Italy *ANTHEM: Advanced Technologies for Human-Centred Medicine*

Scope: Applied ML | GenAI | RAG | Model training | Evaluation | Technical leadership

Impact: Developed language models, retrieval systems and evaluation tools for biomedical research and an Italian news-agency collaboration, while leading researchers and supervising students.

- Led a **four-person team of PhD researchers and postdocs** to develop **11 AI models** addressing a leading news agency's needs in headline generation, semantic search, news categorization and editorial-style text generation. Directed the technical work and personally trained generative language models from scratch on AWS EC2 and HPC infrastructure, fine-tuned models using LoRA and full fine-tuning, and trained semantic retrieval models. Created the common benchmark used to evaluate the team's models, including Italian grammar checks and generation examples.
- Trained and publicly released two Italian-only base language models, **Sophira-1B-base** and **Sophira-360M-base**, using **Megatron-LM on CINECA Leonardo** (64 GPUs and 12 GPUs, respectively). Selected the mixture of public datasets, trained the tokenizer from scratch and chose an architecture suited to small models, balancing training data and model configuration against the available HPC allocation. Evaluated Italian grammatical competence using BLiMP-IT.
- Led the co-development of **2 RAG-based health question answering systems** with MSc students, guiding system design, implementation and evaluation. One system ranked **among the top-performing systems** in the **BioASQ 2026 evaluation campaign**.
- Created **ASPIRE**, an open-source **Streamlit application** deployed for benchmark-based analysis of retrieval performance, helping researchers and students compare systems and identify failure patterns.
- Developed a benchmark for precision medicine, specifically phenotype research, by integrating **12 heterogeneous datasets** into a unified schema with standardized task definitions and evaluation pipelines.
- Mentored and supervised **11 MSc and BSc students** on IR, RAG and applied NLP projects, guiding implementation and evaluation. Taught **Information Retrieval** and **Recommender Systems** through lectures and practical sessions.

Marie Skłodowska-Curie (MSCA) Doctoral Researcher 2020 – 2024
University of Milano-Bicocca, Milano, Italy *EU Horizon 2020 ITN DoSSIER*

Scope: Domain-specific search systems | Biomedical IR | Legal IR | Ranking models | **IR Innovation**

Impact: Conducted my PhD research within the EU Horizon 2020 DoSSIER project, developing retrieval methods and search systems for healthcare and legal domains.

- Proposed and developed **DtMRF**, a decision-theoretic ranking framework combining multiple relevance signals into interpretable, task-specific rankings for professional search. Implemented and applied the framework, evaluated its effectiveness and published the results.

- Designed and implemented **domain-specific search pipelines** for healthcare and legal information access using lexical and neural retrieval models.
- Conducted large-scale experimental evaluation of retrieval architectures across benchmark datasets using ranking metrics and statistical significance testing.
- Collaborated with **international academic and industry teams** across the DoSSIER network, including **Spinque (Netherlands), TU Wien (Austria) and Leiden University (Netherlands)**, on retrieval models, search systems and evaluation methods.
- Delivered multiple research presentations, tutorials, and invited talks at international workshops, conferences, and academic training schools.

Education

PhD in Computer Science (Marie Skłodowska-Curie Fellow) 2020 – 2024
University of Milano-Bicocca, Milano, Italy *Marie Skłodowska-Curie ESR (DoSSIER Project)*

Focus: Domain-specific search models for improving retrieval effectiveness in complex professional search scenarios.
Selected outcomes: WIREs: Data Mining and Knowledge Discovery | Journal of Information Technology & Decision Making

MSc in Advanced Electrical & Computer Engineering 2018 – 2020
Democritus University of Thrace, Xanthi, Greece *Specialization: Information Retrieval*

Focus: Relevance feedback and post-retrieval effectiveness prediction for search ranking systems.
Outcomes published in: Information Retrieval Journal

Integrated Master's in Electrical & Computer Engineering 2013 – 2018
Democritus University of Thrace, Xanthi, Greece *Specialization: Data Mining & Social Media Analytics*

Focus: Real-time social media analytics and decision support for consumer intention monitoring.
Outcomes published in: Operational Research Journal

Selected Research Contributions

29 Publications and research outputs | 212 citations | h-index 9 | Google Scholar

Retrieval Models and Ranking — Implemented DtMRF for multidimensional relevance aggregation; researched query representations, DenseC3 and mixture-of-experts dense retrieval. Publications: **Peikos G.**, Pasi G. ***WIREs DMKD Journal 2024*** | Sokli E., Kasela P., **Peikos G.**, Pasi G. ***WI-IAT 2025 (Best Paper)*** | Sokli E., **Peikos G.**, et al. ***EMNLP 2025*** | DtMRF code

LLM-Enhanced Medical Information Retrieval — Built health-search systems combining BM25, LLM-extracted claims and knowledge-graph checks; evaluated on three TREC Health Misinformation collections. Publications: **Peikos G.**, Kasela P., Pasi G. ***WI-IAT 2024*** | Milanese G., **Peikos G.**, et al. ***ECIR 2025*** | System code

Evaluation and Benchmarking — Created ASPIRE, a deployed Streamlit tool and ECIR system demonstration for comparing retrieval systems and analysing query-level performance. Publications: Kusa W., **Peikos G.**, et al. ***ACM ICTIR 2024*** | **Peikos G.**, Kusa W., et al. ***ECIR 2025 (ASPIRE)*** | Code | Demo | Kusa W., **Peikos G.**, et al. ***BioNLP 2022***

Honors, Awards & Professional Development

Marie Curie Fellowship	Selected for a European Commission Horizon 2020 doctoral fellowship to conduct PhD research within the DoSSIER Innovative Training Network.
Best Paper Award	IEEE/WIC WI-IAT 2025 award for research on mixture-of-experts architectures for dense retrieval systems.
CINECA ISCRA HPC	Principal Investigator, **SpecDLM** (2025): *Innovative Architectures for Small Language Models in Specialized Domains*. Awarded high-performance computing resources through CINECA ISCRA.
Startup Competition Winner	Member of the winning team at the Clean Tech Academy - EIT Deep Tech Talent Initiative for developing an AI-powered circular network supporting electronic device repair and reuse.
Postgraduate Certificate	Postgraduate Certificate in Researcher Professional Development, recognizing advanced researcher competencies across research practice, impact, and professional development.
Grant Proposal School	Training school on transforming research ideas into structured grant proposals for European research funding.

Academic Service

Academic Reviewing	Reviewer and Program Committee member for 22 international conferences and journals in Information Retrieval, NLP, and AI systems. Reviewer for national and international research grant proposals.
Community & Training	Organizer of workshops and hackathons; delivered tutorials and invited talks at international conferences and training schools.